



Steel Coil Defect Prediction

Surface defect prediction using PLC sensor data and machine learning analytics.

- **PRESENTED BY**
DSBA Team
- **PROGRAM**
ML Fieldwork

CSI: Coil Scene Investigation

00





Introduction

01

- Steel production generates large volumes of sensor data across the manufacturing process
- Defects significantly impact product quality, costs, and efficiency
- Traditional inspection methods are reactive and limited
- Machine Learning enables data-driven defect prediction using historical production data
- By combining PLC (**programmable logic controller**) sensor data with defect records, we can identify patterns and root causes
- This project focuses on building a predictive model to detect defects early and support real-time decision-making



Project overview

01

Objective

- identify root causes of defects in metal coil production
- Use PLC sensor data and defect inspection records



Approach

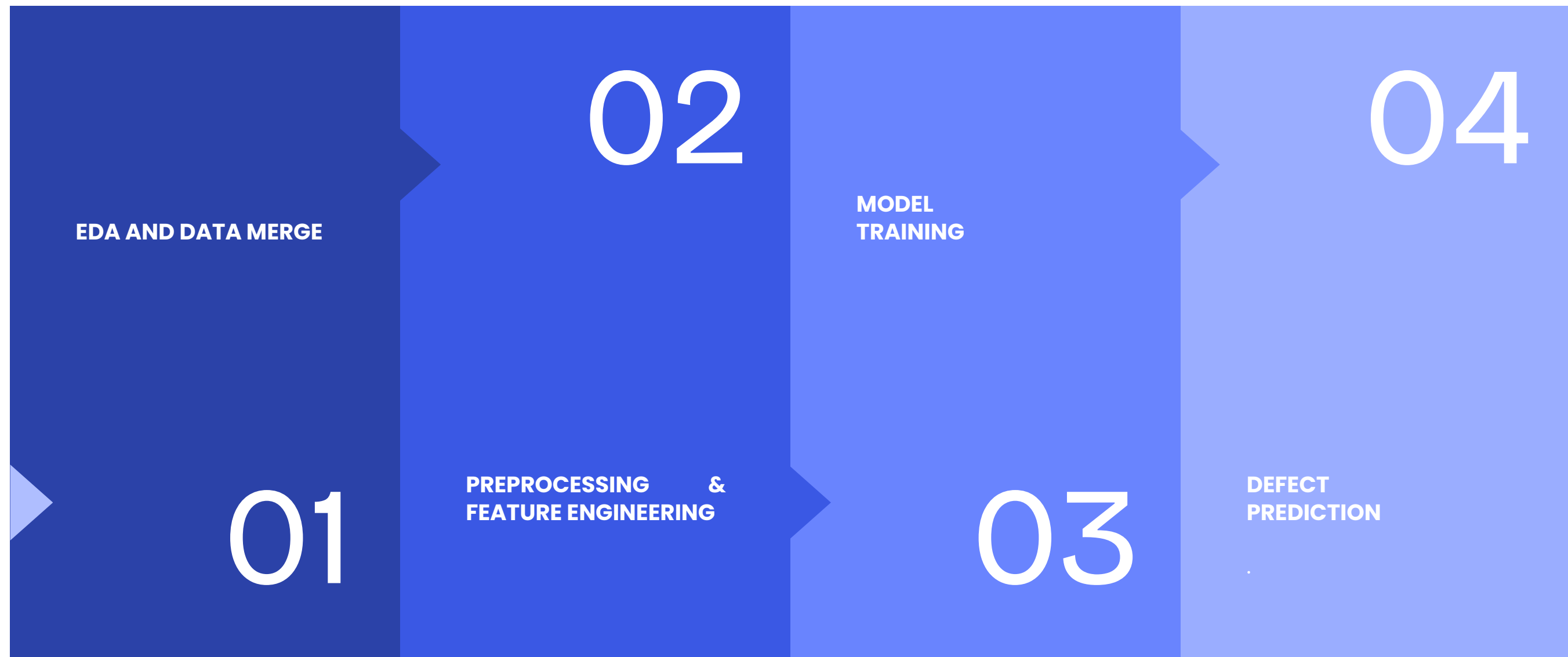
- Combine production sensor data with defect labels
- Apply machine learning models for prediction

Business goal

- Reduce waste and production costs
- Improve process control and product quality

Methodology

MACHINE LEARNING PIPELINE FOR DEFECT PREDICTION[↗]



EDA PLC Sensor & Defects Data

PLC Sensor Data

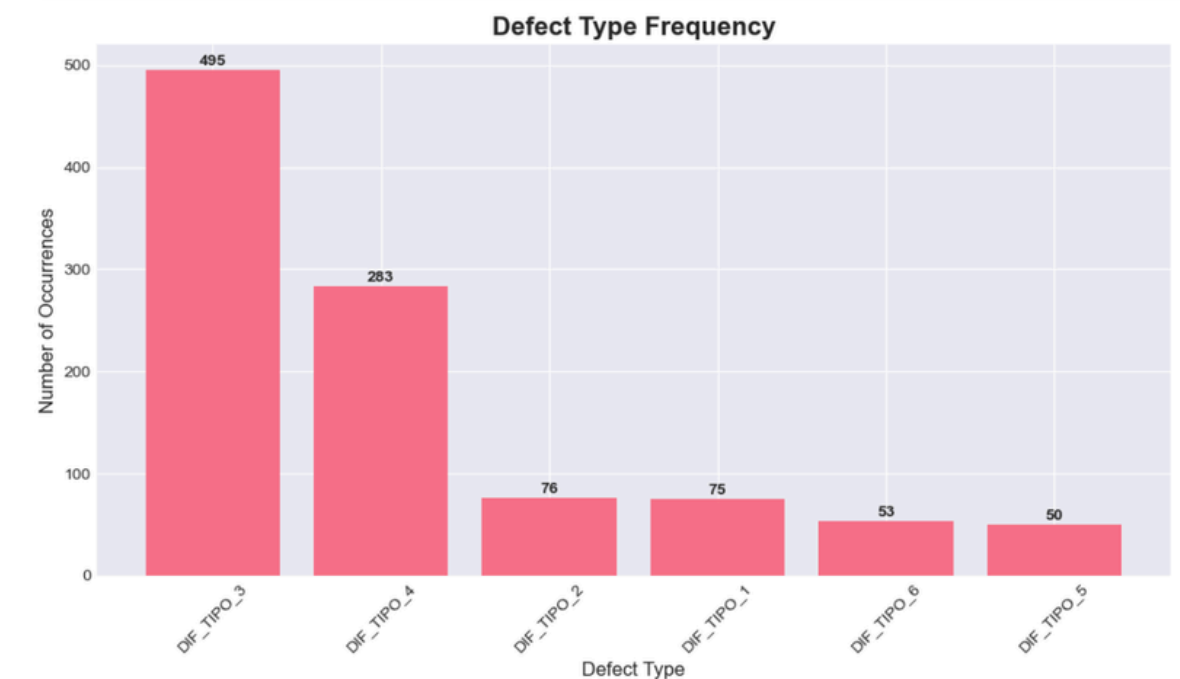
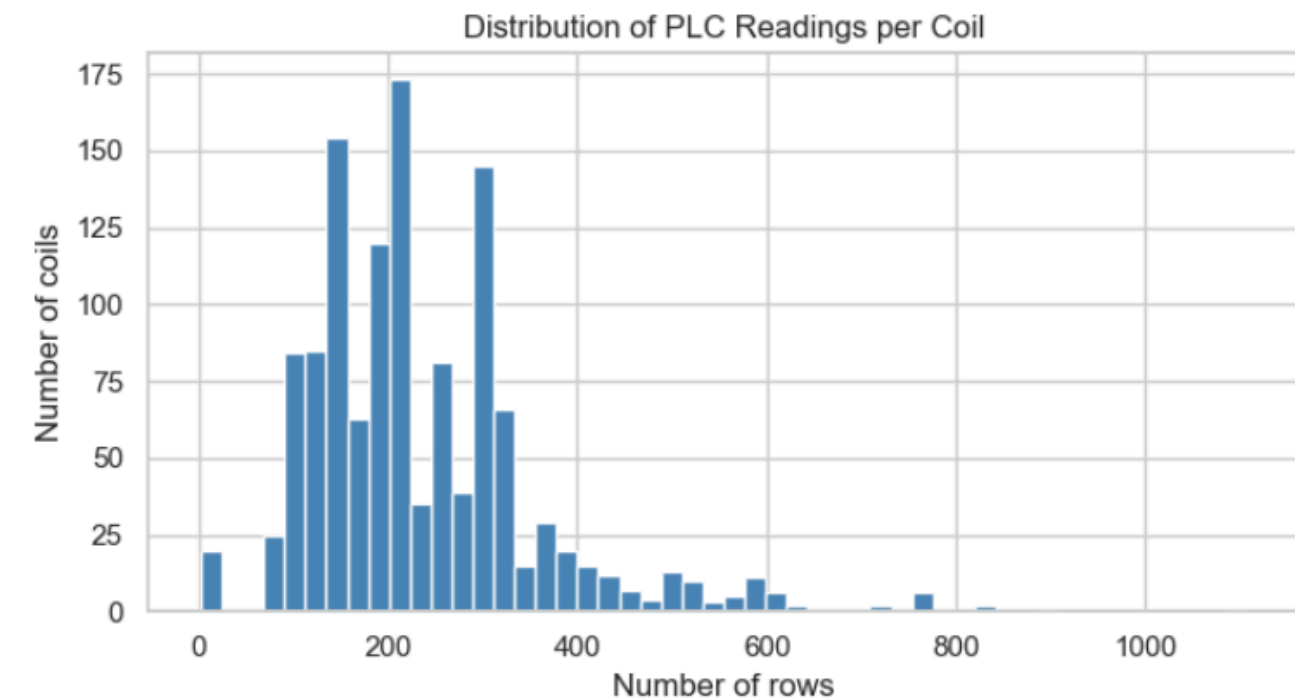
- **299,384 rows**: one reading every 7 metres per coil
- **1,261 unique coils**: March to May 2019
- **106 sensor** channels across 9 production sections
- Key columns: **COIL, MT (metre position), DATE**

Defect Label Data

- **617 rows**: one per defect segment
- **6 binary defect types**: Tipo 1 through Tipo 6
- Each row covers a metre range (MT_FROM to MT_TO)
- No data dictionary: physical meaning of each Tipo **unknown**

Key challenge to highlight:

- two files, two different languages — exact **positions vs ranges**
- **74% of defect segments** cover more than **80% of coil length**



Scan for the process

The Merge Strategy

01

The Problem

1. PLC speaks in exact positions — metre 483
2. Defect file speaks in ranges — metre 482 to 1539
3. Simple join on COIL alone would flag every row as defective — wrong

The solution: position range join

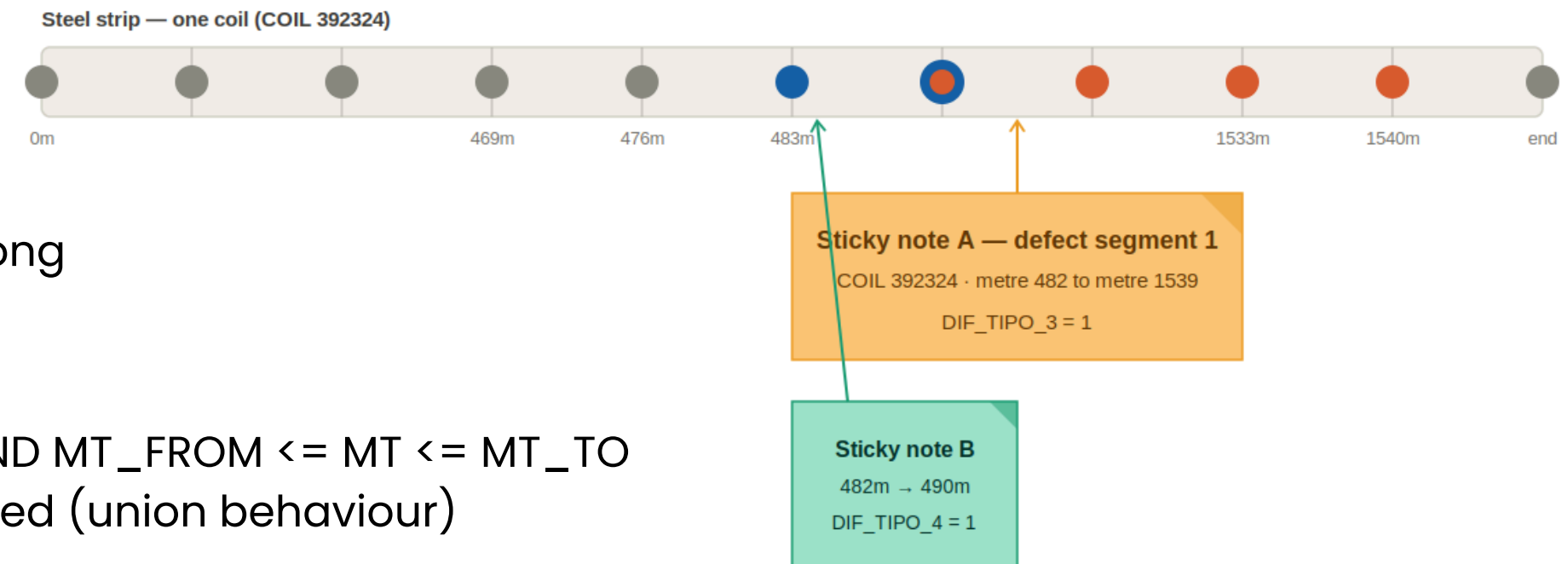
- For each defect segment find all PLC rows where COIL matches AND $MT_FROM \leq MT \leq MT_TO$
- Never overwrite a 1 with a 0 — overlapping segments both preserved (union behaviour)
- Rows with no match stay 0 — clean strip

```
Final dataset (flagged coils removed):
Before removing flagged : 299,384 rows
After removing flagged   : 298,851 rows
Coils removed            : 21
Defective rows           : 106,526 (35.6%)
Clean rows               : 192,325 (64.4%)
Final shape              : (298851, 120)
```

✓ Saved to: ../data/cleaned/plc_defect_merged.csv

Merge strategy — position range join

How PLC sensor rows get matched to defect segments



Key rule — never overwrite a 1 with a 0

When two sticky notes overlap on the same row — both defect flags are preserved. This is the union behaviour.

Legend

- Grey — outside any defect range → all flags = 0 (clean row)
- Blue — inside sticky note B only → DIF_TIPO_4 = 1
- Orange — inside sticky note A only → DIF_TIPO_3 = 1
- Double — inside BOTH notes → TIPO_3 = 1 AND TIPO_4 = 1

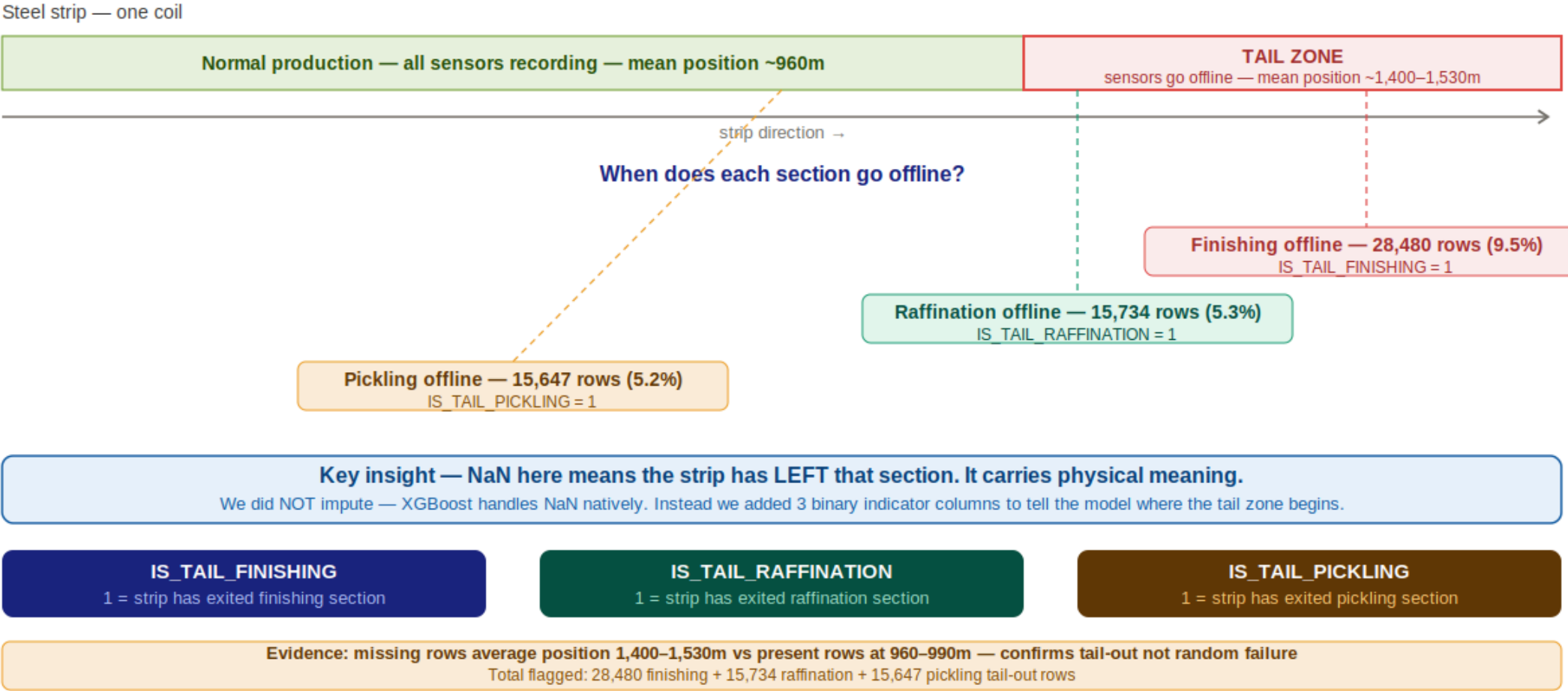
Result after merge

MT=476m → TIPO_3=0, TIPO_4=0, HAS_DEFECT=0 (clean) | MT=483m → TIPO_3=1, TIPO_4=1, HAS_DEFECT=1 (both defects preserved)

EDA Key findings

OVERVIEW OF PLC SENSOR DATA COLLECTION AND QUALITY METRICS

Finding	What	Action
1.Short coils	19 coils with 10 rows or fewer	Flagged: later excluded in later step
2.Tail-out effect	Missing data at coil end — strip exits each section	Added 3 binary indicator columns
3.AIR_CH4 negatives	Sensor noise at burner shutdown — physically impossible	Clipped to zero
4.No stuck sensors	All 60 sensors have meaningful variance	No action needed
5.Correlations	128 correlated pairs — all physically explained	All 60 sensors kept
6. Process stability	Median CV = 0.039 — very stable process	2 unstable coils flagged



Not Enough for a detective : What does the 6 tipos mean

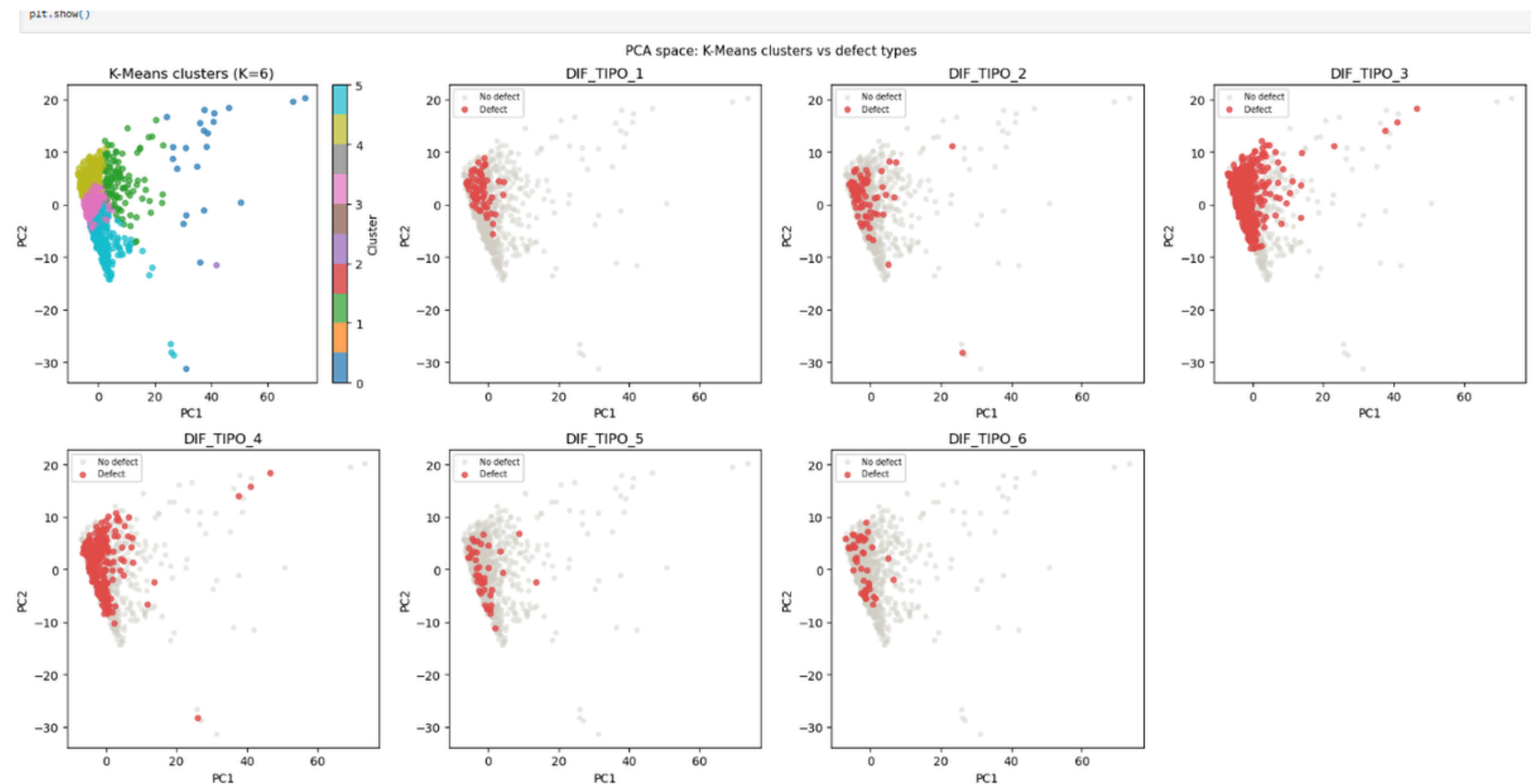
01

Unsupervised clustering (K-Means, K=6)

on coil-level sensor aggregates to reverse-engineer the physical meaning of each defect label: since no data dictionary existed

Method:

- Aggregate by mean & std per sensor per coil = removes autocorrelation.
- Scale using StandardScaler = put all sensors on equal footing, which prevents temperature dominating.
- Reduce: PCA (212 to 35 components) Removes correlated sensor pairs (12* found in EDA) = reduce noise.
- Cluster K-Means K=6 [K=6 to match the defect counts], HDBSCAN tested but version conflict
- Validate : cross- reference cluster vs DIF_TIPO 1-6. Defect flags used only AFTER clustering, never as a feature.



Tipo 1 – Tipo 6 appear in the same region of PCA space. If the defect labels mapped to distinct process conditions, each chart would look different. They don't, confirms our clustering finding = labels were assigned by visual inspection, not derived from sensor data

Not Enough for a detective : What does the 6 tipos mean

02

Key Findings:

- Defect types doesn't map directly to a distinct sensor condition. Each cluster contain a mix of defect types, suggesting labels were assigned by visual inspection rather than derived from process data
- The client independently verified this finding, defect propagation introduces noise into the sensor-label relationship.

Defect Label Data Dictionary					
Physical meaning reverse-engineered from K-Means clustering — no data dictionary existed					
Tipo	Prevalence	Confidence	Sensor signature	Hypothesis	Top cluster
Tipo 3	34.6% most common	Medium	High raffination laser + low pre-heat pressure + elevated zone 5	Scale/oxidation driven by raffination strip position and insufficient pre-heating	Cluster 4 (50%) highest risk
Tipo 4	19.9%	Medium	Almost always co-occurs with Tipo 3 — same sensor signature	Same root cause as Tipo 3 presenting differently on visual inspection	Cluster 4 (30%)
Tipo 1	5.3%	Low	High raffination laser + elevated zone 5 combustion	Strip running high post-furnace under elevated zone 5 conditions	Cluster 4 (10%)
Tipo 2	5.7%	Low	High zone 1+2 furnace pressure + lower finishing force	Pressure-related defect from over-pressurised early furnace zones	Cluster 1 (7%)
Tipo 5	3.3% — rare	Very low	No dominant sensor pattern identified	No clear process driver may depend on material properties or external factors	Spread across clusters 1,3,5
Tipo 6	3.6% — rare	Very low	Weak association with high raffination laser + zone 5 pattern	Loosely associated with same conditions as Tipo 3 and 4	Cluster 4 (5%) very weak

Confidence reflects data availability — Tipo 3 and 4 are the most actionable findings
Tipo 5 and 6 require domain expert confirmation — insufficient coils for strong signal

What the data DID tell us: two process drivers

02

The finding:

- Clusters separate coils by **defect severity** not **defect type**
- Cluster 4: 407 coils: has **50% Tipo 3 and 30% Tipo 4**, double the average defect rate
- **Two sensors** consistently distinguish high-risk from low-risk clusters.

The two drivers:

- **LASER_RAFF_1**: strip position post-furnace: When the strip exits the furnace at the wrong height: surface defects are more likely.
- **AIR_Z5 / GAS_Z5**: zone 5 furnace combustion; When zone 5 runs too hot, the strip surface is damaged before cooling even begins.



*"When the strip exits at the wrong position AND zone 5 is running at a higher temperature than usual, defect risk doubles. These two sensors are the **earliest warning signs** in the entire production line. Confirmed independently by both **clustering and SHAP analysis**"*

Affected Coils



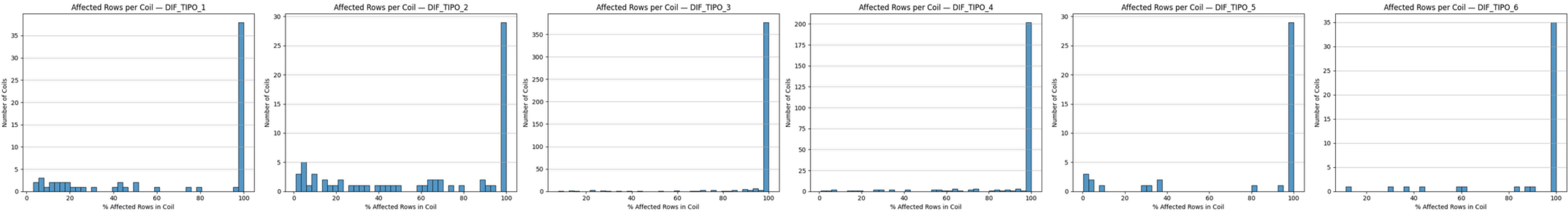
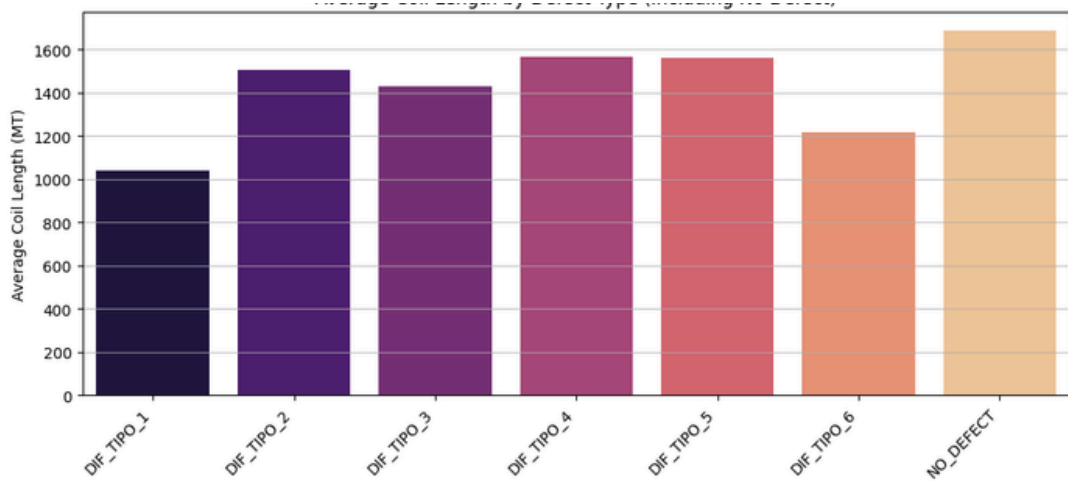
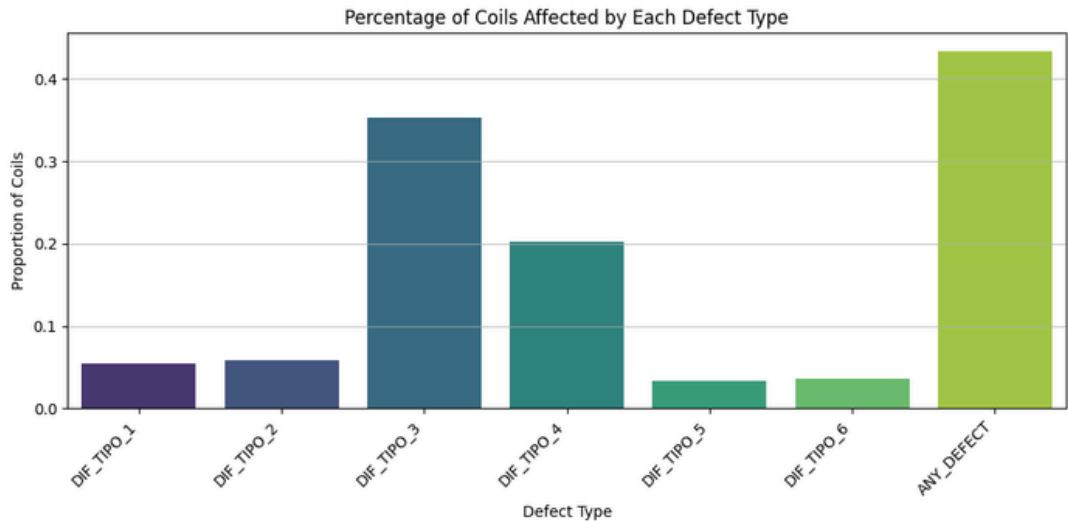
Average Length

Number of Coils per Affected %

Defects 3
and 4

OUR MAIN TARGETS

- SHORT AND GAPPY COILS
HAVE ALREADY BEEN
REMOVED
- **SIMILAR AVERAGE LENGTH**
BETWEEN COILS WITH AND
WITHOUT DEFECTS



Features Reduction

120 Initial Features

10 Semantic Groups, 73 Total Features

36 Remaining Features

PCA within each
Semantic Group

SelectKBest to preserve the
Interpretability of the
Features

30 Features after PCA
95% of Variance Retained per Group

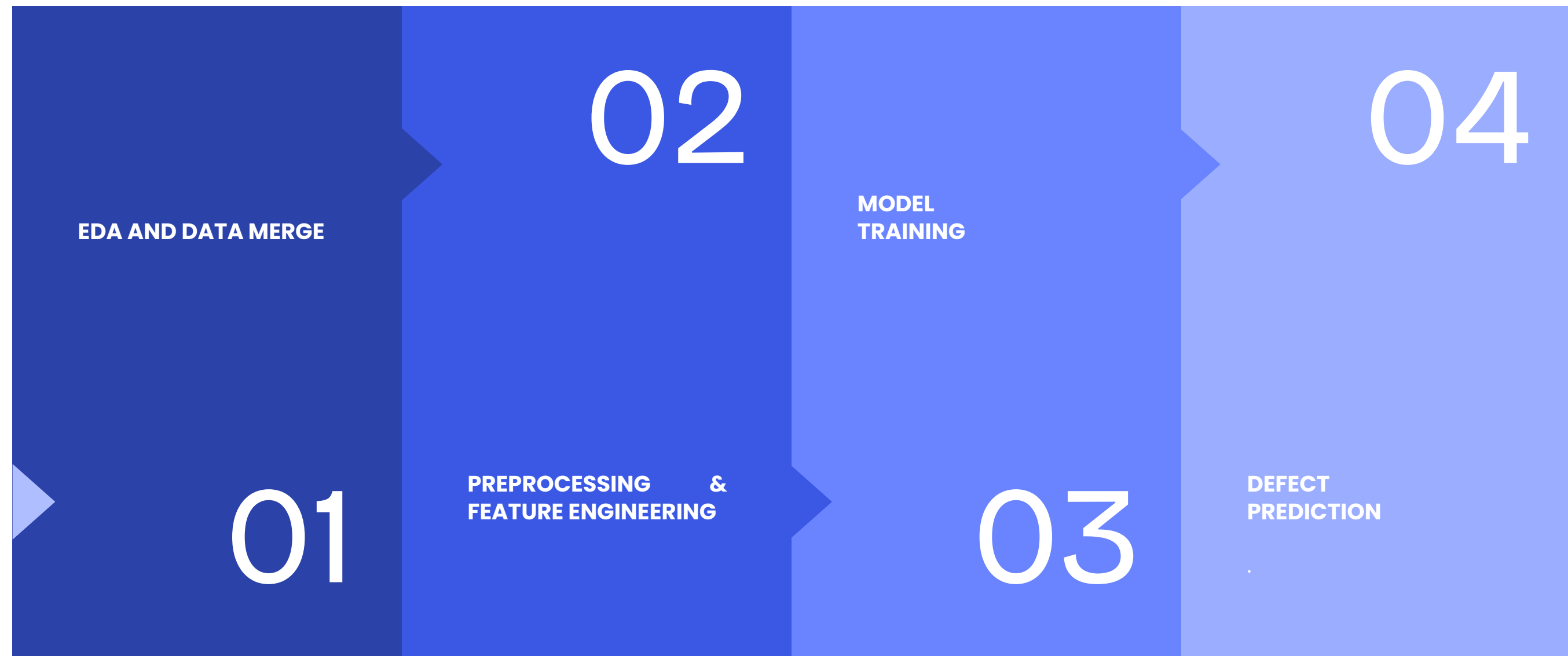
29 Features after SelectKBest
80.56% of Variance Retained

59 Final Features

Temperature_Zones,
Pyrometers,
Laser_Furnace,
Air_Channels,
Air_Zones,
Gas_Zones,
Pressure_Zones,
Ventilation,
Cooling,
Laser_Cooling

Methodology

MACHINE LEARNING PIPELINE FOR DEFECT PREDICTION[↗]



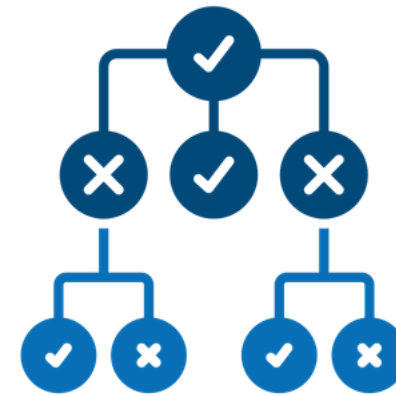
The Model Pipeline

XGBoost



GroupKFold

Hyperparameter Tuning

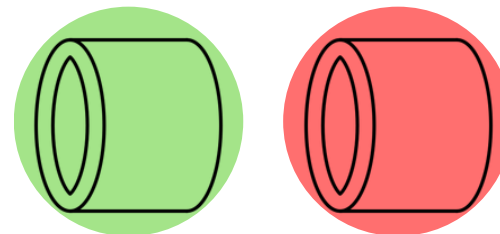


- **DECISION-TREE BASED MODEL:** AN ENSEMBLE OF WEAK DECISION TREES BUILT SEQUENTIALLY, WITH EACH TREE CORRECTING ERRORS OF PREVIOUS ONES

- HANDLES WELL **CLASS IMBALANCE**

- **ROBUST TO CORRELATED FEATURES**

- **GOOD INTERPRETABILITY** (IMPORTANCE AND SHAP)



THE DATA WAS SPLIT BY COIL INTO TRAIN AND TEST IN ORDER TO **AVOID DATA LEAKAGE**

colsample_bytree : 0.8
 learning_rate : 0.1
 max_depth : 5
 n_estimators : 150
 reg_alpha : 0.0
 reg_lambda : 0.5
 subsample : 0.8

- **EACH TREE** RANDOMLY RELIES ON AT LEAST THE **80%** OF THE TRANSFORMED **FEATURES**

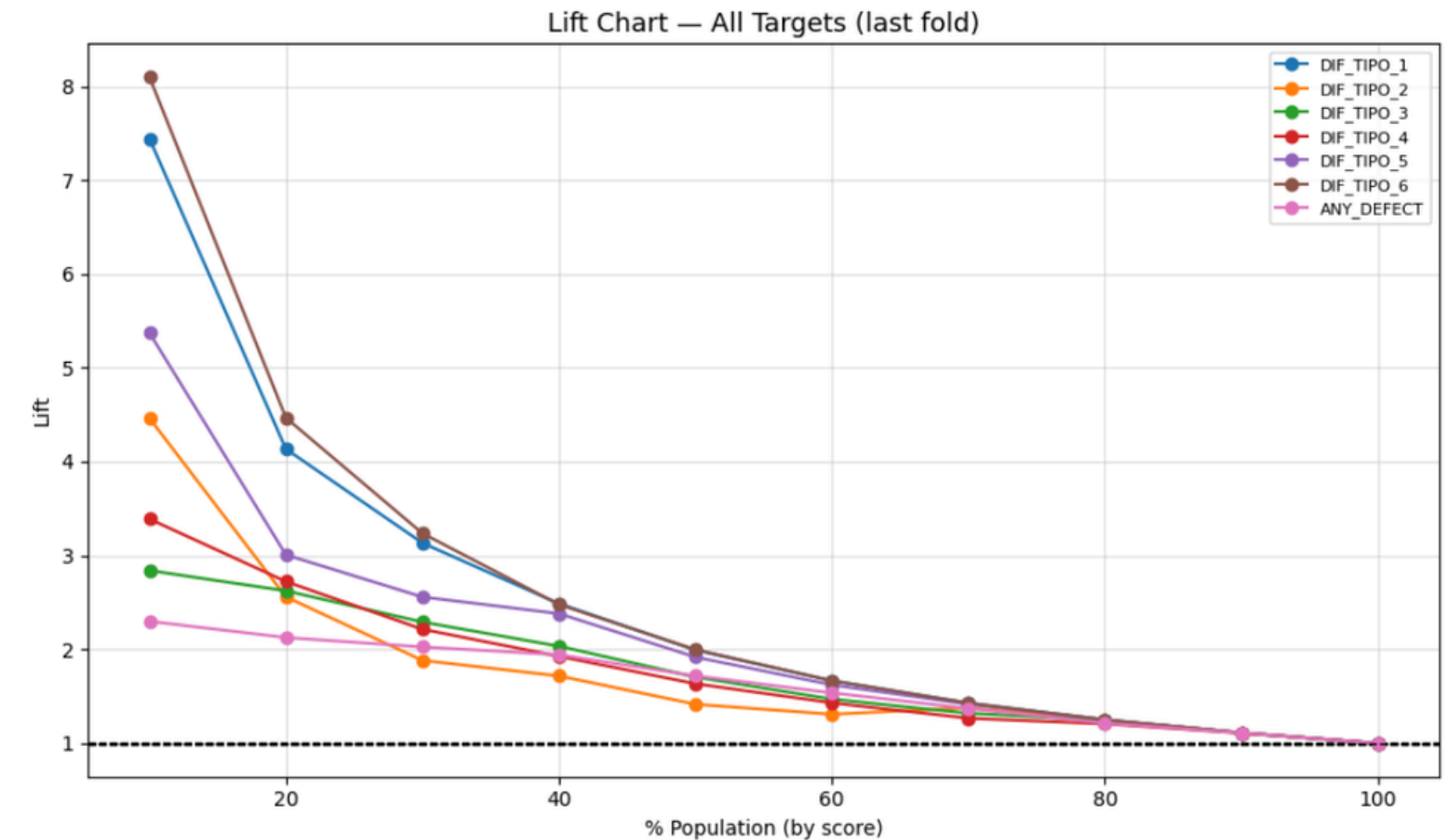
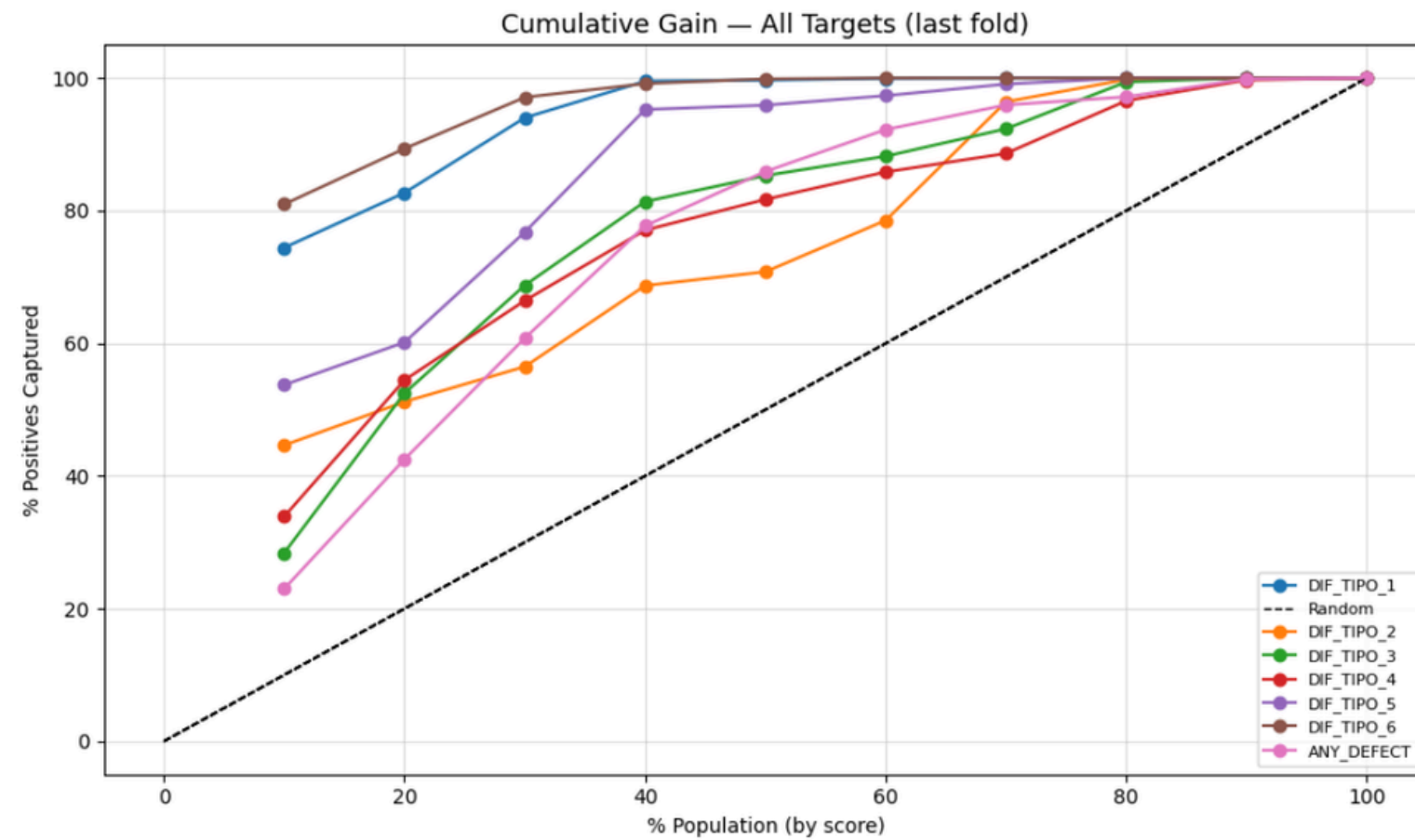
- **CAUTIOUS LEARNING PROCESS:** ONLY THE 10% (**0.1 LEARNING_RATE**) OF CONTRIBUTION OF EACH TREE AFFECTS THE ENSEMBLE

- **150 TREES** SEQUENTIALLY BUILD

Overall Performance

03

Target	Mean Accuracy	Std Accuracy	Mean ROC-AUC	Std ROC-AUC	Lift @top-20%	Gain @top-30%
DIF_TIPO_1	0.9794	0.0045	0.9115	0.0529	4.1350	94.0000
DIF_TIPO_2	0.9581	0.0151	0.7144	0.0827	2.5600	56.5000
DIF_TIPO_3	0.8309	0.0126	0.8867	0.0106	2.6240	68.8000
DIF_TIPO_4	0.8523	0.0123	0.8318	0.0265	2.7230	66.4000
DIF_TIPO_5	0.9700	0.0072	0.7724	0.1206	3.0080	76.8000
DIF_TIPO_6	0.9829	0.0032	0.9640	0.0233	4.4680	97.1000
ANY_DEFECT	0.8114	0.0096	0.8814	0.0170	2.1260	60.8000



THE GAIN CHART : ORDER THE COILS IN A DESCENDING WAY BY THE SCORE PREDICTED BY THE MODEL. FOR DEFECTS 3 AND 4, **IN THE FIRST 20% OF THE COILS , THE MODEL CAPTURES ALMOST 60% OF THE DEFECTIVE POPULATION**, AND GOES TO 80% FOR THE FIRST 40% OF THE COILS.

THE LIFT CHART : FOR DEFECTS 3 AND 4, **IN THE FIRST 20% OF THE COILS , THE MODEL PREDICTS THE DEFECTS ALMOST 3 TIMES BETTER THAN RANDOM GUESSING.**

Performance
on Main Targets

Defect_3

35% OF THE COILS

83.1%

MEAN PREDICTION
ACCURACY

— DIF_TIP0_3 —	precision	recall	f1-score	support
No Defect	0.85	0.89	0.87	38495
Defect	0.75	0.66	0.70	18145
accuracy			0.82	56640
macro avg	0.80	0.78	0.79	56640
weighted avg	0.82	0.82	0.82	56640

Defect_4

20% OF THE COILS

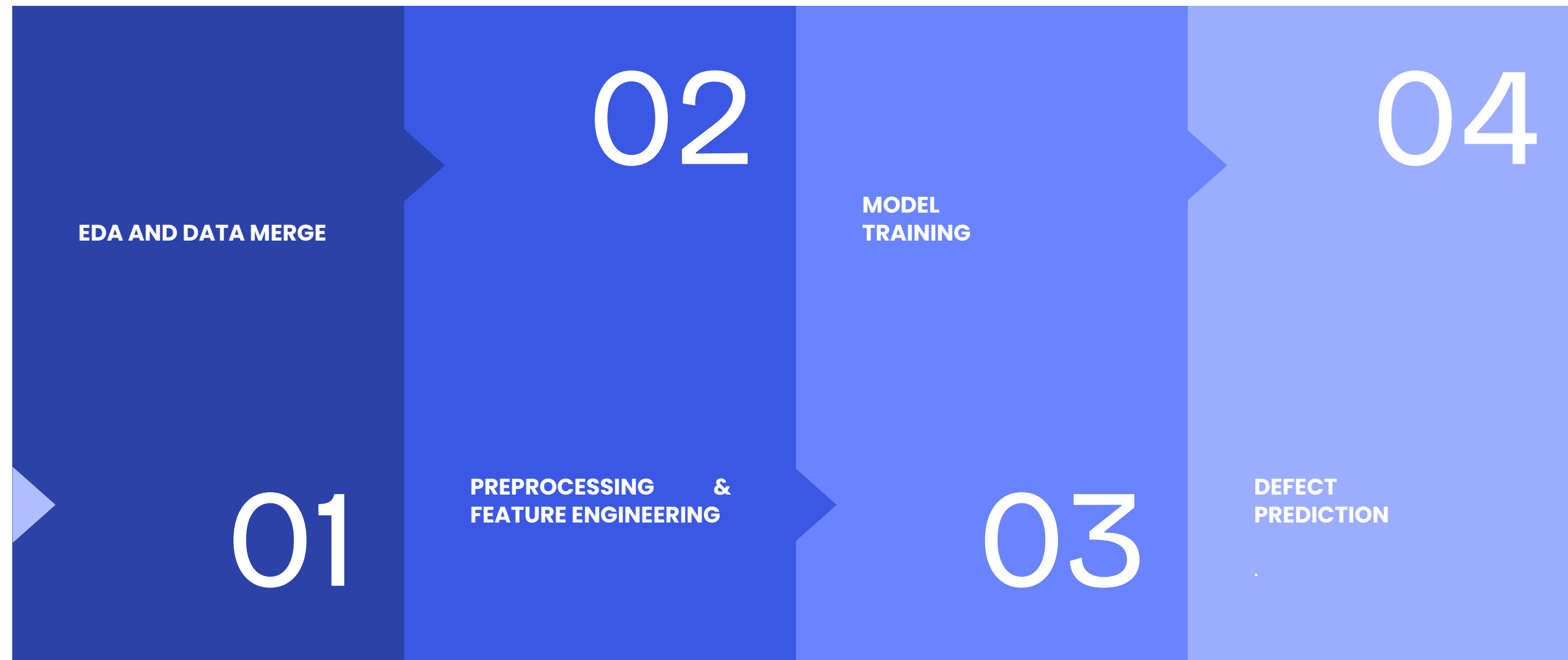
85.2%

MEAN PREDICTION
ACCURACY

— DIF_TIP0_4 —	precision	recall	f1-score	support
No Defect	0.86	0.95	0.90	44490
Defect	0.68	0.43	0.53	12150
accuracy			0.83	56640
macro avg	0.77	0.69	0.71	56640
weighted avg	0.82	0.83	0.82	56640

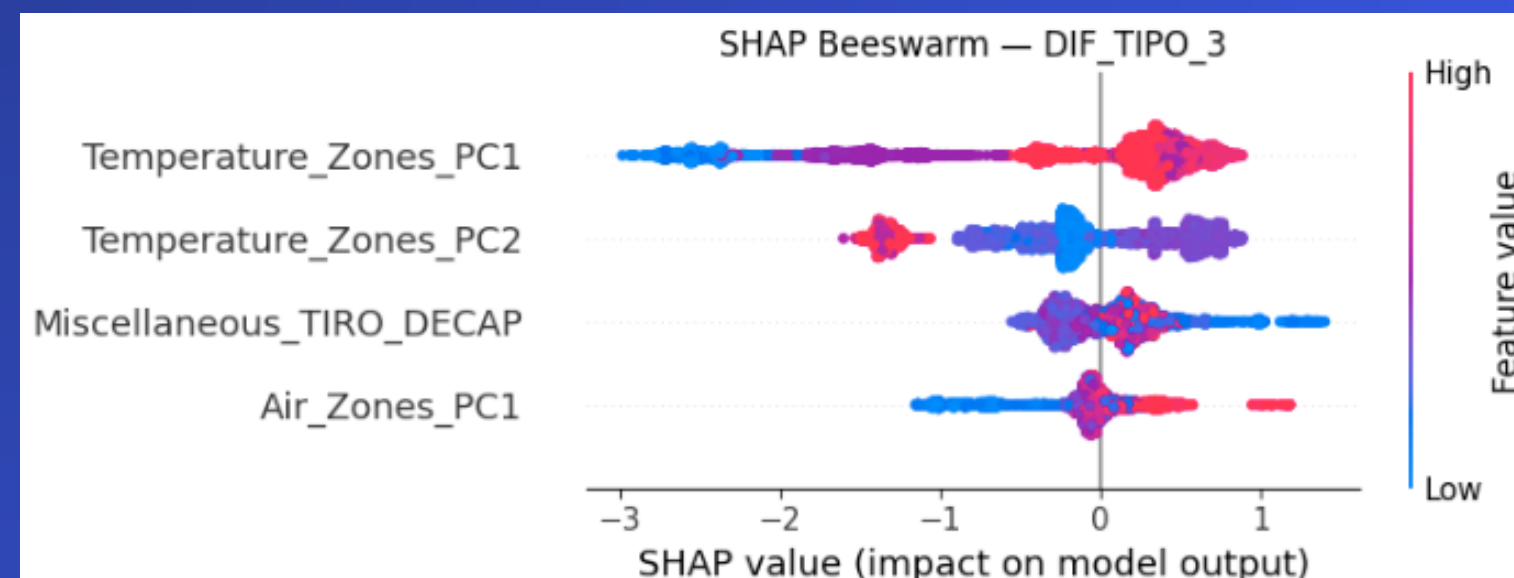
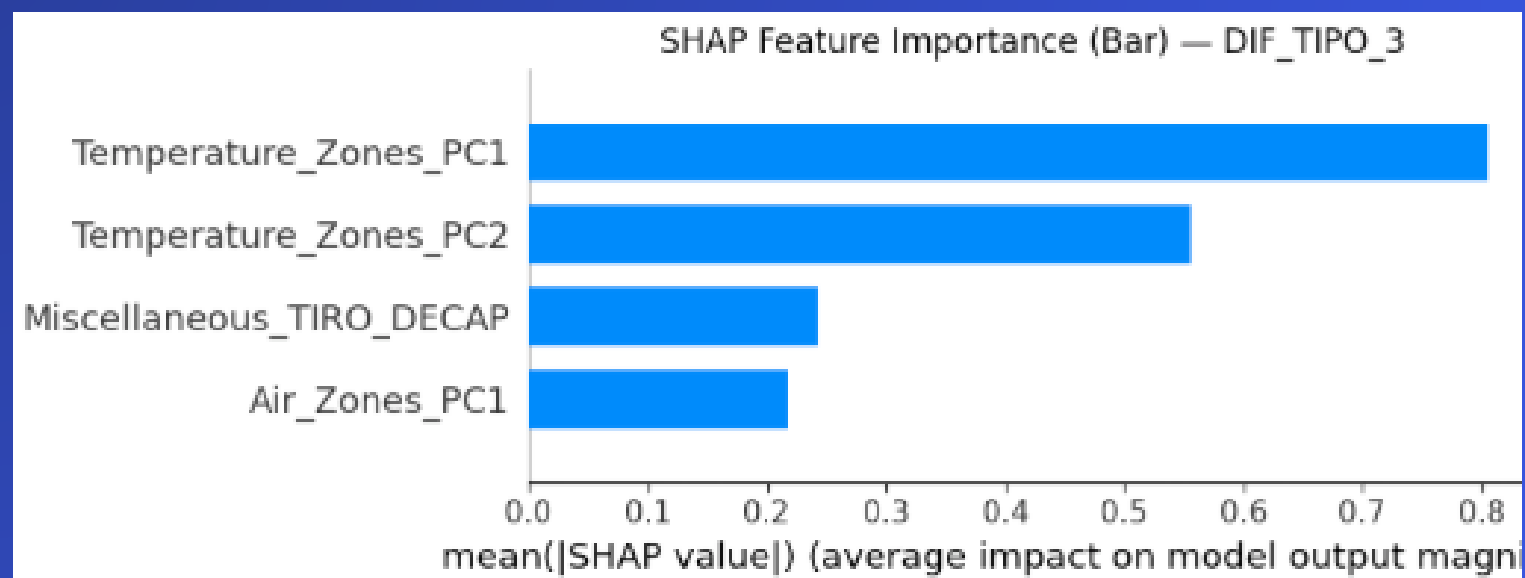
Methodology

MACHINE LEARNING PIPELINE FOR DEFECT PREDICTION[↗]



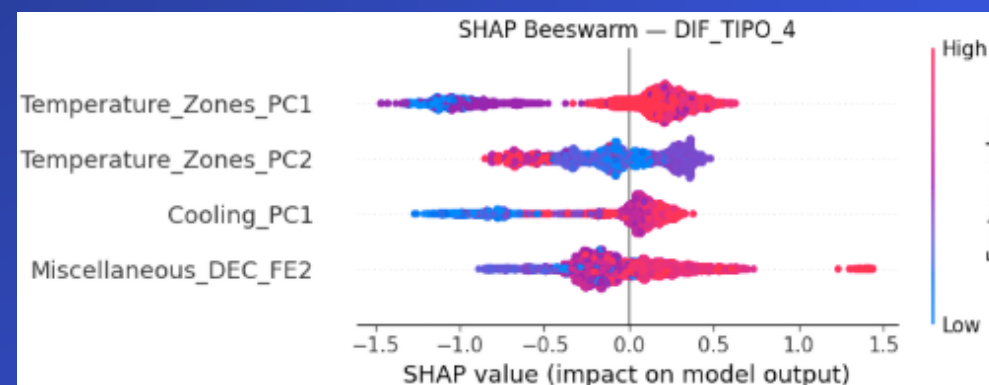
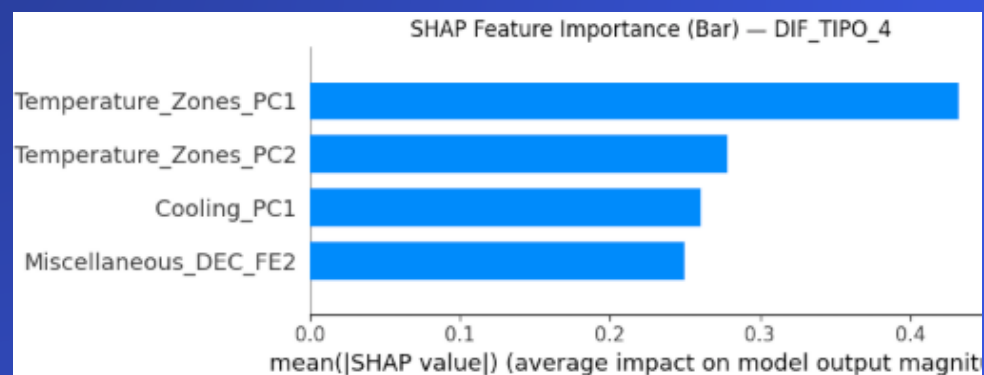
Key Insights

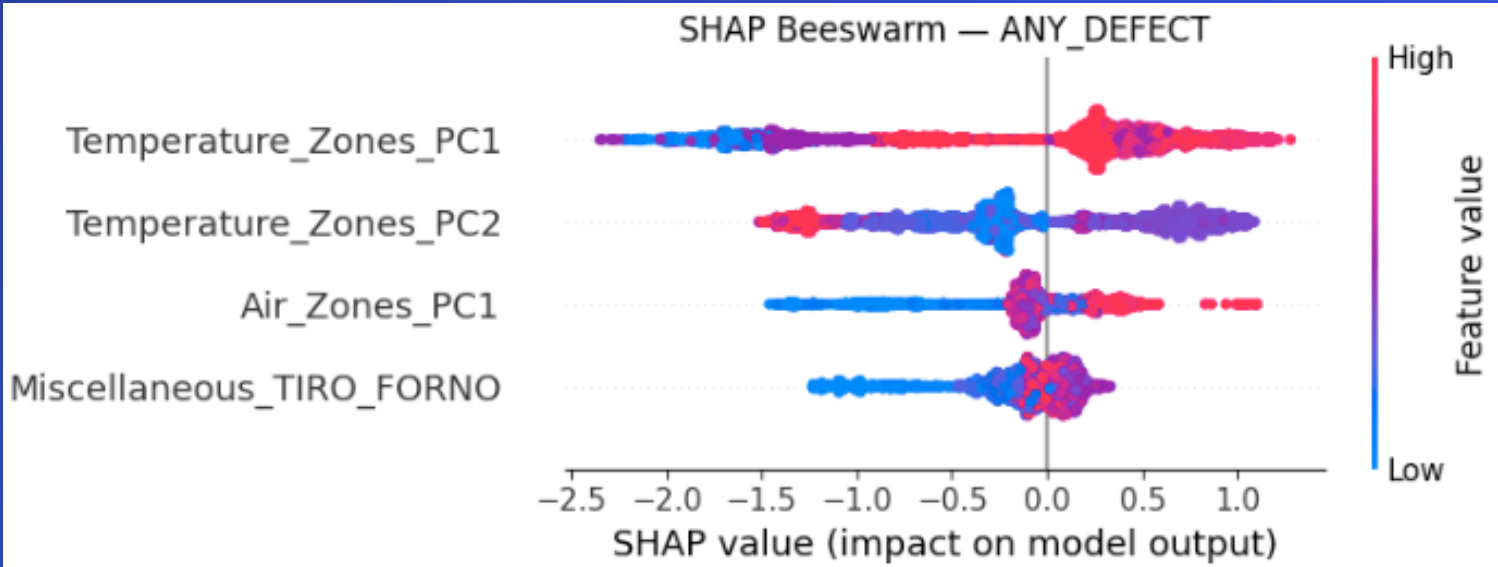
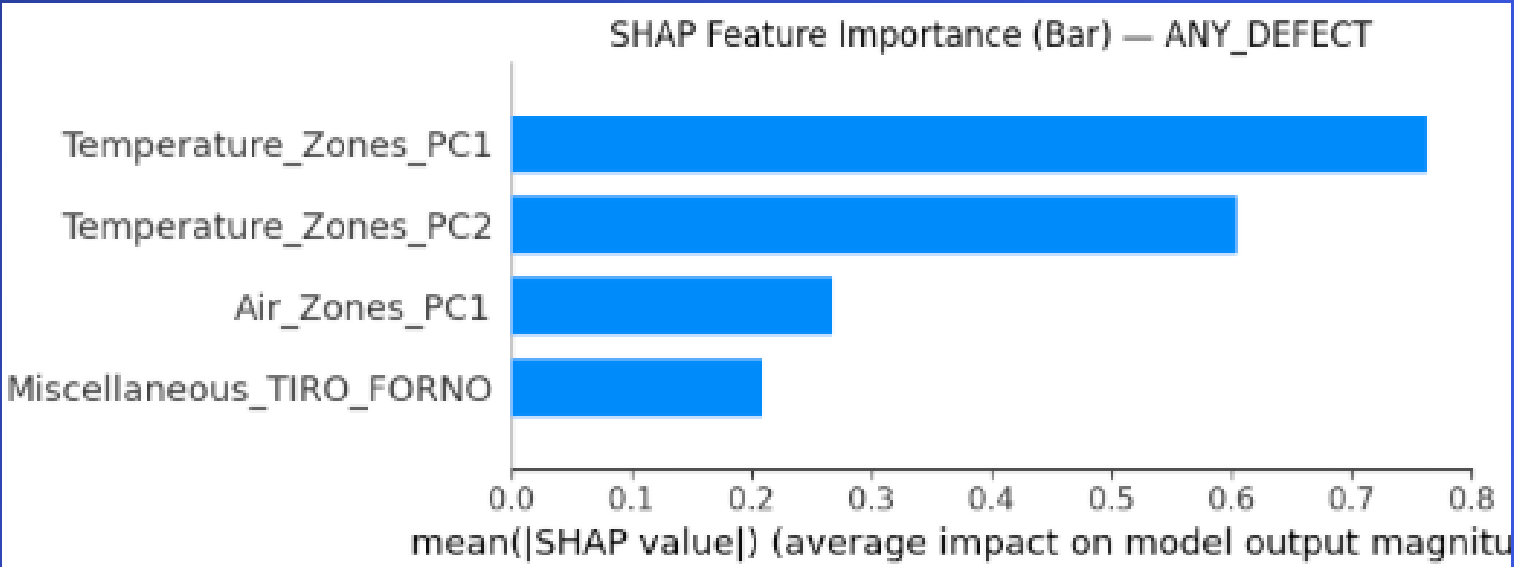
04



Temperature is the most significant factor for defect prediction

For positive values, the probability of a defect increases.
For negative values, the probability of a defect decreases.
The colour indicates the intensity of the feature value detected by the sensor





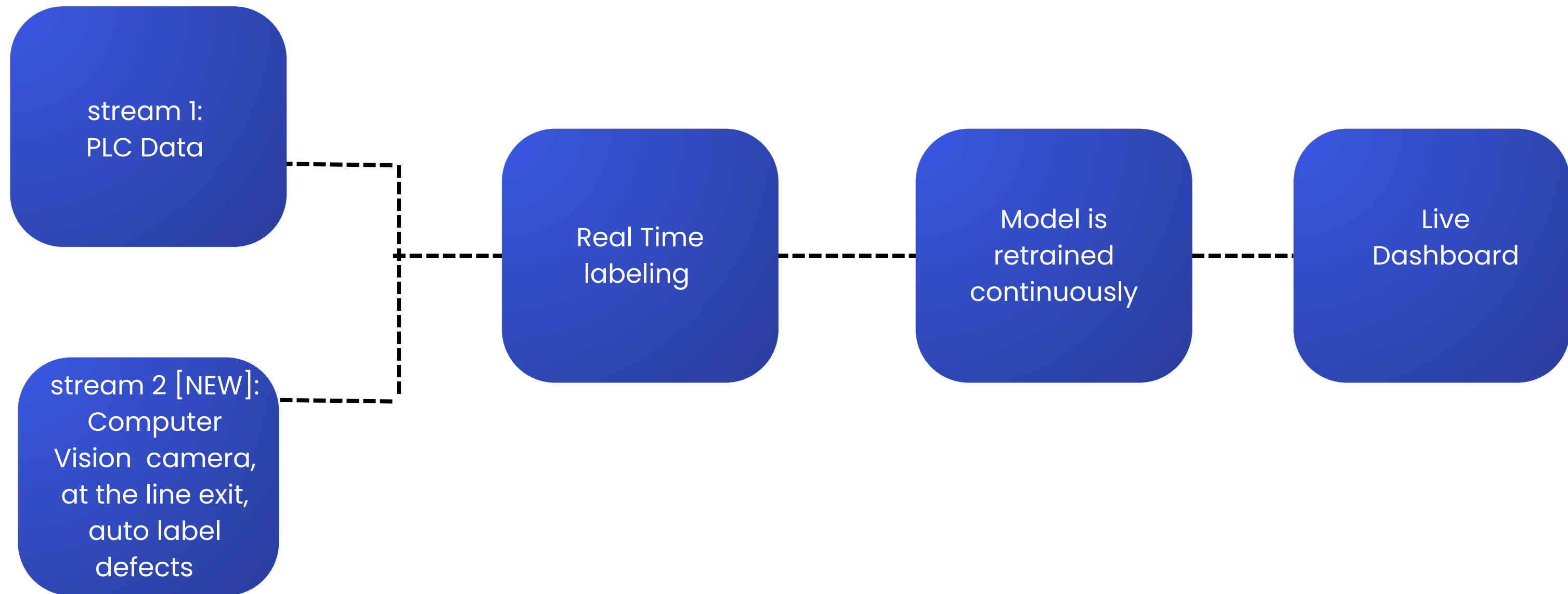
— ANY_DEFECT —				
	precision	recall	f1-score	support
No Defect	0.84	0.86	0.85	34532
Defect	0.77	0.74	0.76	22108
accuracy			0.81	56640
macro avg	0.81	0.80	0.80	56640
weighted avg	0.81	0.81	0.81	56640

Temperature throughout the entire process remains the most significant feature in predicting defects.

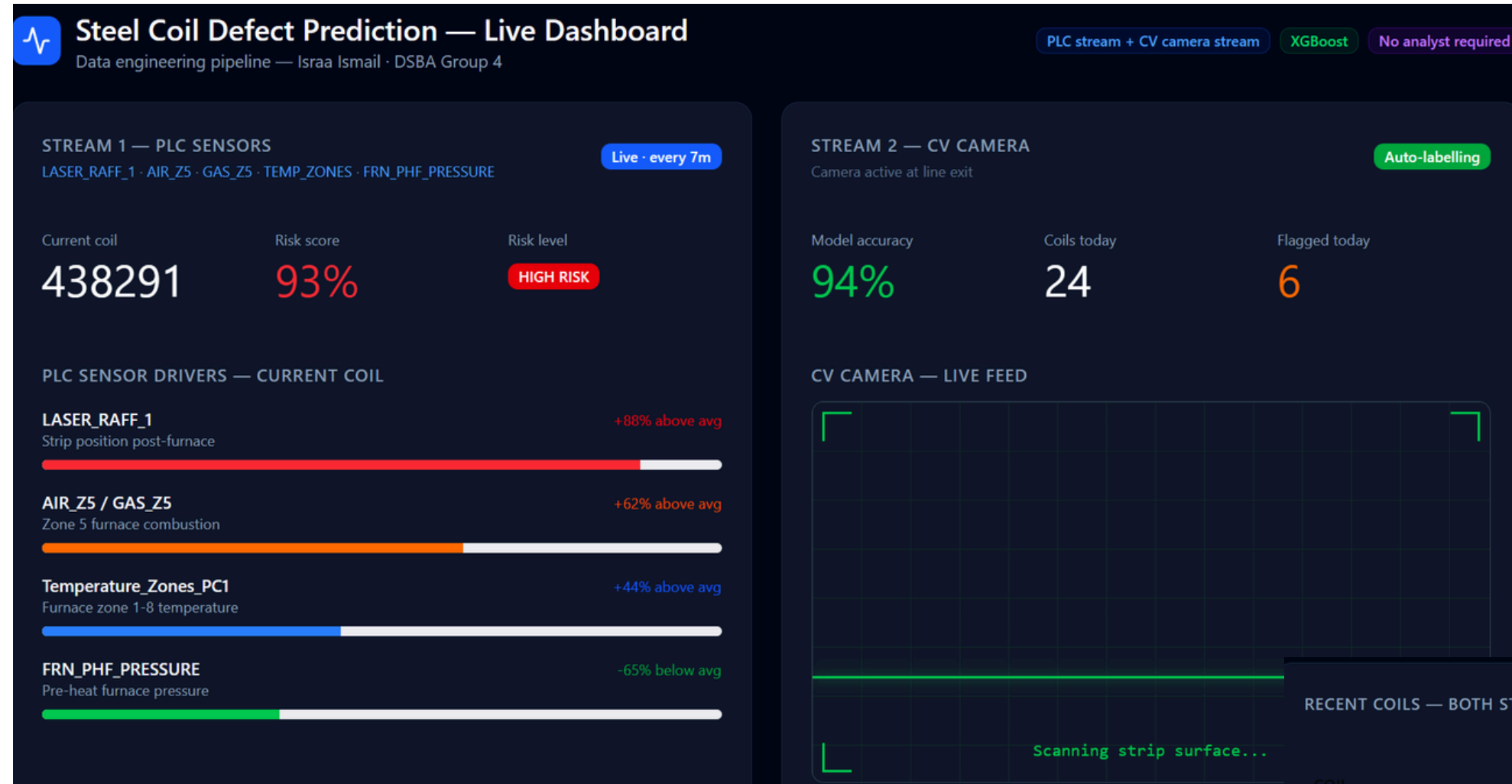
Stabilising temperatures and preventing thermal shocks can play an important role in predicting and reducing defects in coils productions.

The Solution from the detectives

RE-DESIGN THE CURRENT PROCESS



The Solution from the detectives



Recent coils — both streams			
Coil	PLC Score	CV Label	Status
438291	87%	Defect	High
438290	22%	Clean	Clear
438289	61%	Defect	Medium
438288	18%	Clean	Clear
438287	91%	Defect	High

Thank You!

https://drive.google.com/file/d/1qHfA_rPaDwuH8GAy1c2CgQqBw4oNUZLH/view?usp=sharing

<https://github.com/Ilsmail12/ML-Project/tree/main>



Extras

XGBOOST AND GROUPOKFOLD

